README

Project Links

Project Overview

Main Features

How to Use the Web App

1. Use Image Extraction
2. Calculate the Log-rank Test
3. Use Sample Data
4. Use the Indirect Comparison Tool

Data Format

Technical Workflow

Project Strengths

Limitations

Project Links

Web application:

https://kaplan-meier-curve-llm-extraction-log.onrender.com/

GitHub repository:

https://github.com/TongW1029/Kaplan-Meier-Curve-LLM-extraction-Log-rank-Analysis

(Note: The API key has been embedded, so there is no need to enter the API key.)

Project Overview

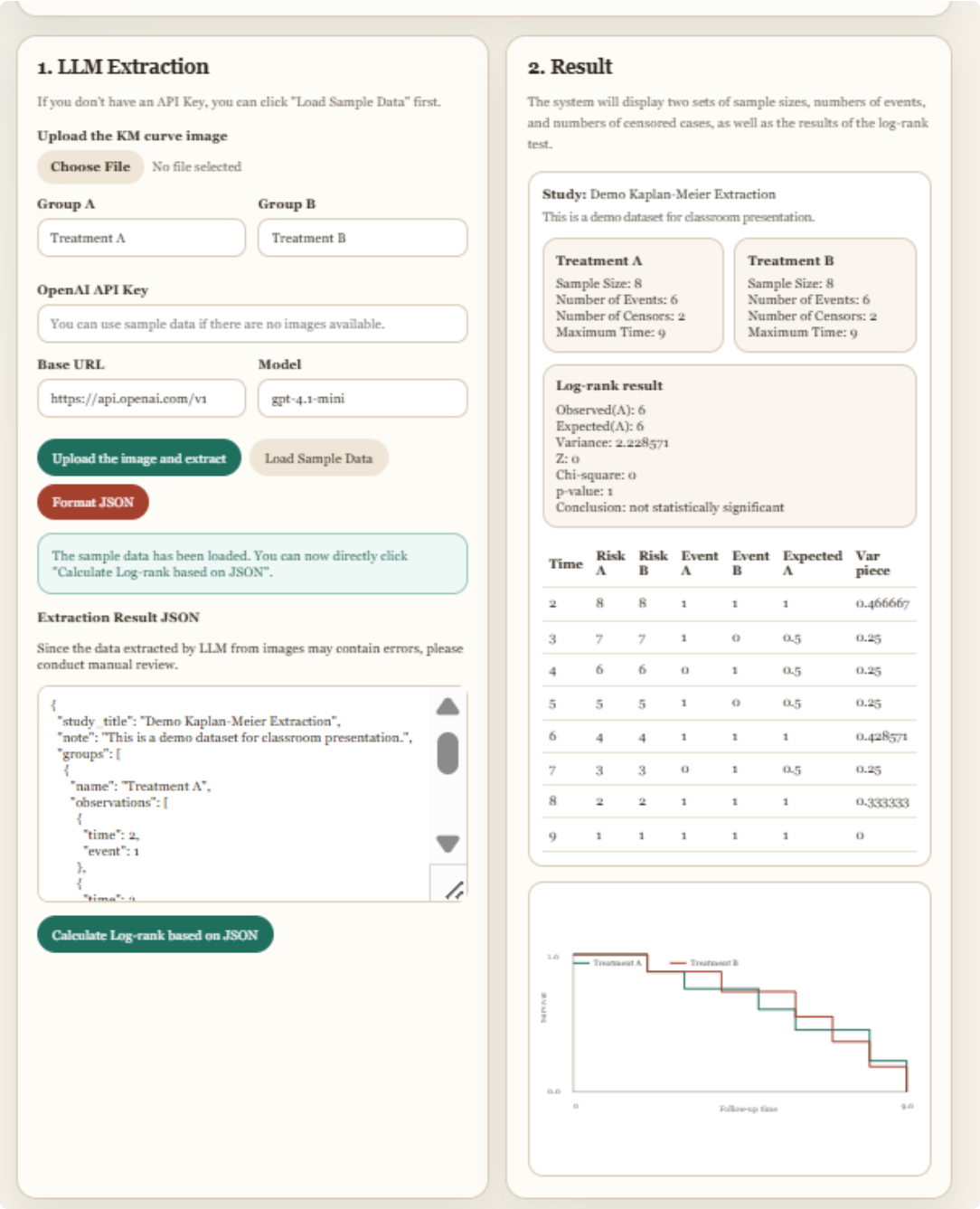
This project is a web-based tool for a survival analysis course project. The goal isAIM to extract approximate survival data from a Kaplan-Meier curve image and then calculate a log-rank test based on the extracted data.

Users can upload a Kaplan-Meier curve image. The web app uses an LLM API to identify the two survival curves and generate approximate `time` and `event` data. Then, the system

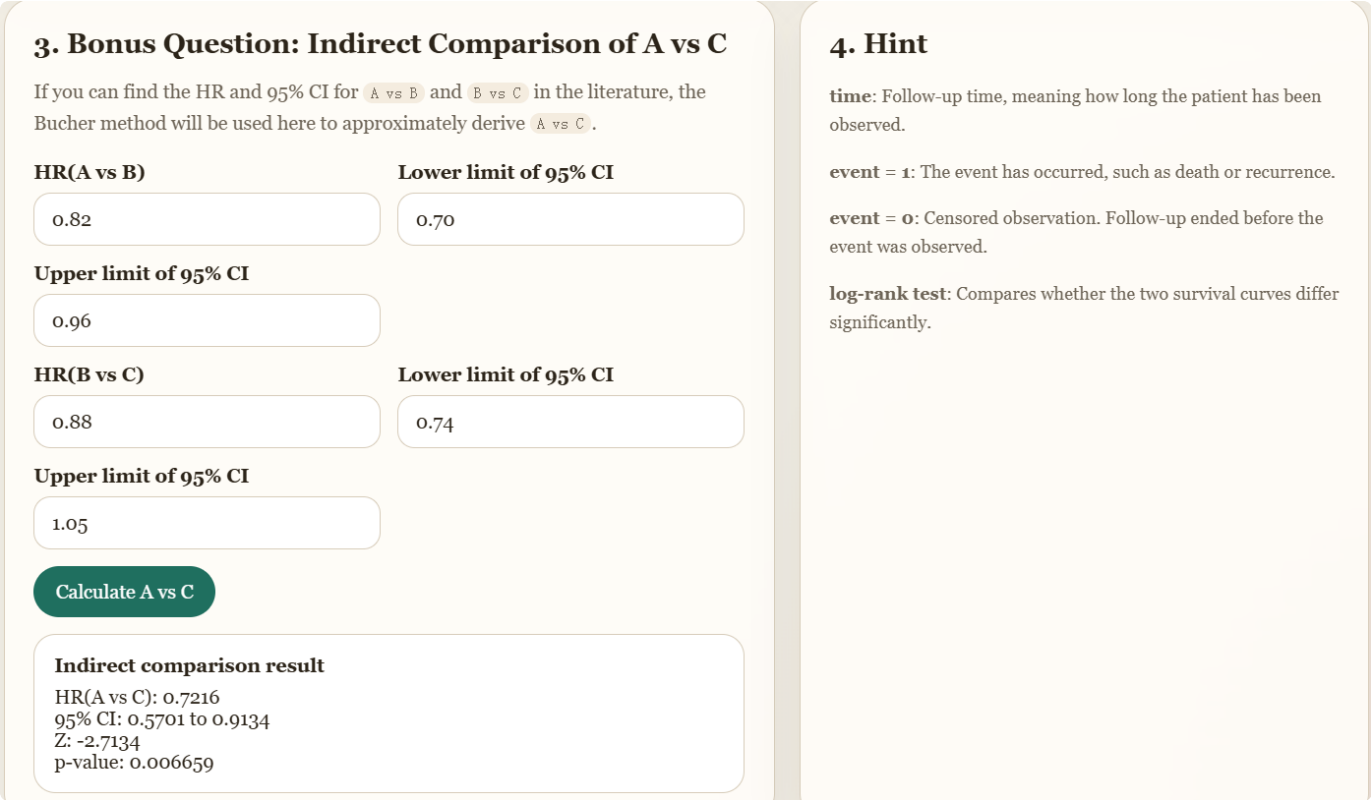
automatically calculates the log-rank test and displays the sample size, number of events, number of censored observations, p-value, and an approximate survival curve plot.

Main Features

1. Upload a Kaplan-Meier curve image
2. Use an LLM API to extract approximate survival data from the image
3. Generate `time` and `event` data in JSON format
4. Calculate the log-rank test based on the extracted data
5. Display sample size, events, censored observations, and p-value
6. Plot an approximate Kaplan-Meier survival curve



7. Provide an indirect comparison tool for A vs C as an additional feature

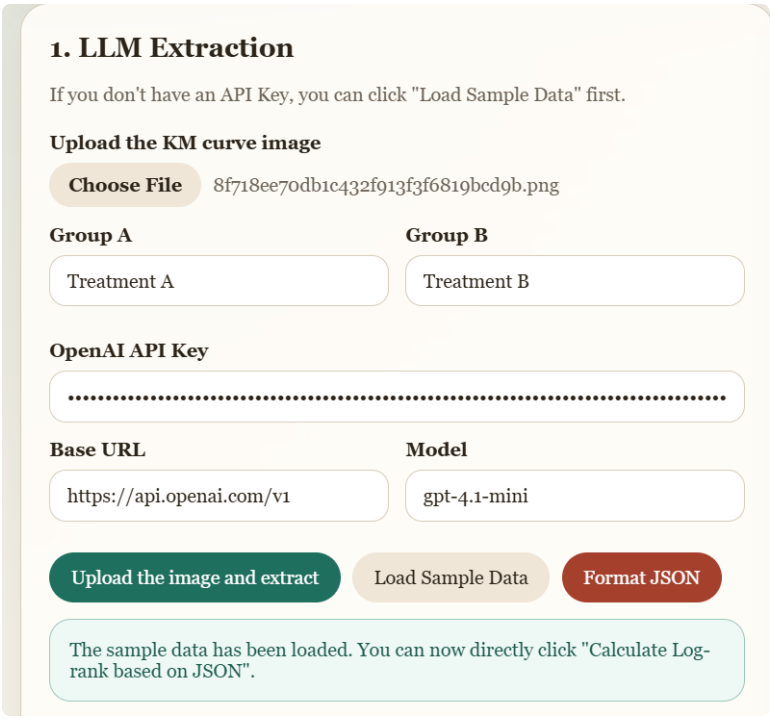


How to Use the Web App

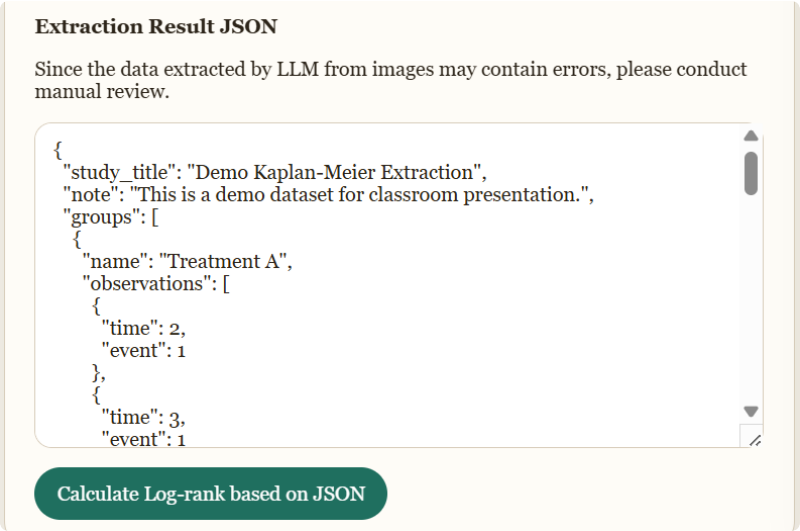
1. Use Image Extraction

In the LLM Extraction section:

1. Click `Choose File` to upload a Kaplan–Meier curve image
2. Enter the two group names, such as `Treatment A` and `Treatment B`
3. Enter an OpenAI API Key
4. Click `Upload the image and extract`
5. Wait for the extracted result



The extracted data will be shown in the JSON box.



2. Calculate the Log-rank Test

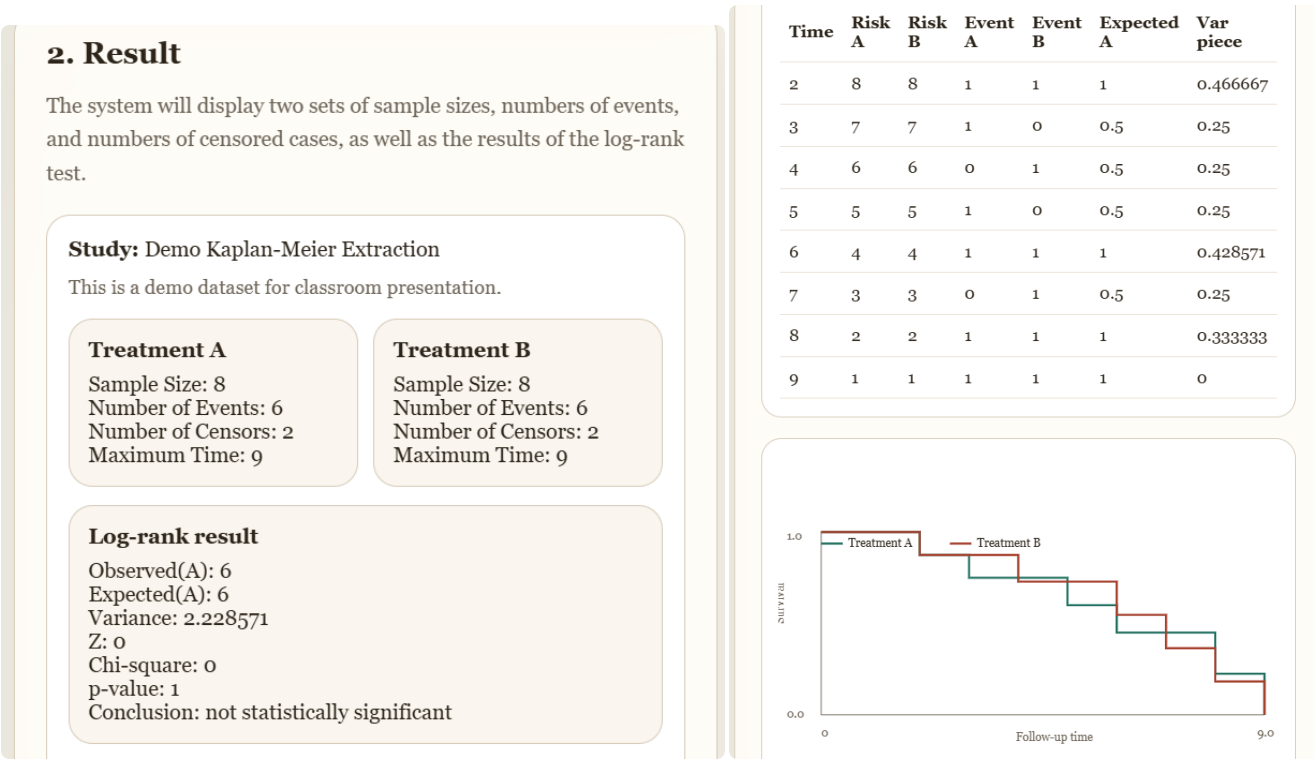
After the JSON data is generated, click:

`Calculate Log-rank based on JSON`

The results section will show:

- Sample size for each group
- Number of events for each group
- Number of censored observations for each group

- Log–rank test p–value
- Approximate survival curve plot



2. Result

The system will display two sets of sample sizes, numbers of events, and numbers of censored cases, as well as the results of the log-rank test.

Study: Demo Kaplan-Meier Extraction

This is a demo dataset for classroom presentation.

Treatment A
Sample Size: 8
Number of Events: 6
Number of Censors: 2
Maximum Time: 9

Treatment B
Sample Size: 8
Number of Events: 6
Number of Censors: 2
Maximum Time: 9

Log-rank result
Observed(A): 6
Expected(A): 6
Variance: 2.228571
Z: 0
Chi-square: 0
p-value: 1
Conclusion: not statistically significant

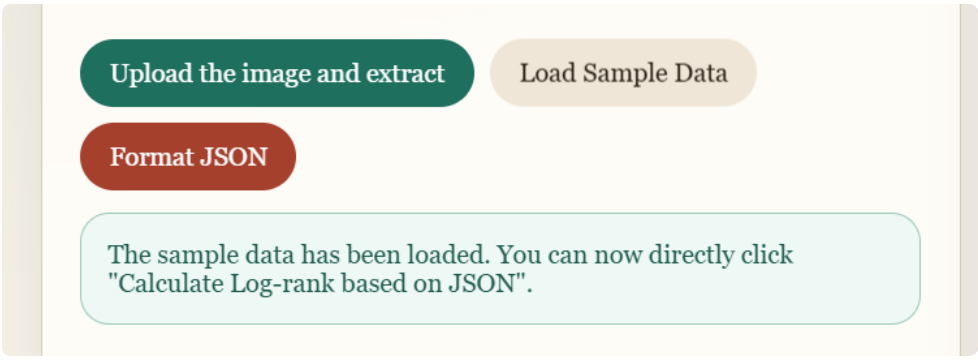
Time	Risk A	Risk B	Event A	Event B	Expected A	Var piece
2	8	8	1	1	1	0.466667
3	7	7	1	0	0.5	0.25
4	6	6	0	1	0.5	0.25
5	5	5	1	0	0.5	0.25
6	4	4	1	1	1	0.428571
7	3	3	0	1	0.5	0.25
8	2	2	1	1	1	0.333333
9	1	1	1	1	1	0

3. Use Sample Data

If an API Key is not available, users can click:

`Load Sample Data`

This loads example data and allows users to test the log–rank calculation and result display.



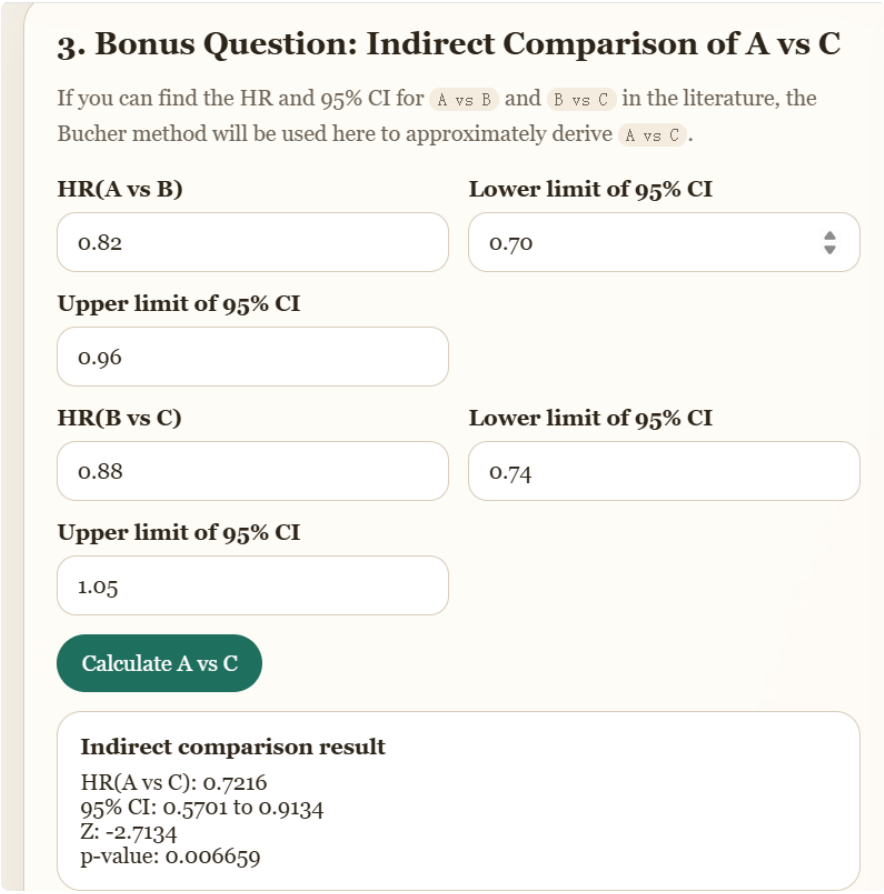
4. Use the Indirect Comparison Tool

The bottom section of the page includes a bonus tool for indirect comparison. If the user has literature values for:

- HR(A vs B)

- HR(B vs C)
- 95% confidence intervals for both HRs

The tool can estimate the indirect comparison result for A vs C.



Data Format

The core data format is JSON. Each patient-level record contains two variables:

```Plain Text
{
  "time": 6,
  "event": 1
}
```

Explanation:

- `time` means follow-up time
- `event = 1` means the event occurred
- `event = 0` means the observation was censored

Technical Workflow

The basic workflow of this project is:

```plain text
Kaplan-Meier curve image
-> LLM API extracts approximate time/event data
-> User reviews the JSON
-> Backend calculates the log-rank test
-> Frontend displays statistical results and curves
```

The frontend is built with HTML, CSS, and JavaScript. The backend is built with Python. The backend receives the uploaded image, calls the LLM API, processes the JSON data, and calculates the log-rank test.

Project Strengths

1. Combines LLM image understanding with survival analysis
2. Allows users to upload Kaplan-Meier curve images directly
3. Keeps the extracted JSON visible so users can review and edit it
4. Displays both statistical results and an approximate survival curve
5. Includes sample data for demonstration

Limitations

Reconstructing patient-level `time` and `event` data from a Kaplan-Meier curve image is an approximate process. The LLM extraction may be affected by image quality, curve complexity, legend clarity, and whether a number-at-risk table is available.

Therefore, this project is best used as a course project and method demonstration tool. It should not be considered a formal clinical research software tool. Users should review and, if necessary, manually correct the JSON output generated by the LLM.